The electron – releasing alkyl group R pushes electron towards nitrogen in the amine ($R\text{-}\ddot{N}H_2$) and provide unshared electron pair more available for sharing with proton.

Therefore, the expected order of basicity of aliphatic amines (in gas phase) is

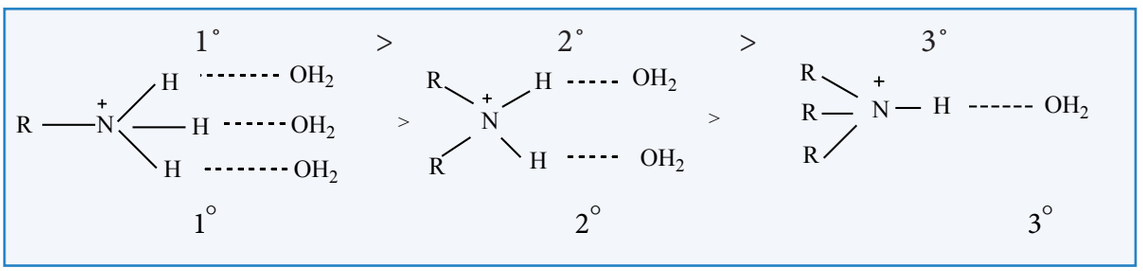
$$\underset{(3^0)}{R_3\ddot{N}} > \underset{(2^0)}{R_2\ddot{N}H} > \underset{(1^0)}{R\text{-}\ddot{N}H_2}$$

The above order is not regular in their aqueous solution as evident by their pKb values given in the table.

To compare the basicity of amines, the inductive effect, solvation effect, steric hindrance, etc., should be taken into consideration.

Solvation effect

In the aqueous solution, the substituted ammonium cations get stabilized not only by electron releasing (+I) effect of the alkyl group but also by solvation with water molecules. The greater the size of the ion, lesser will be the solvation. The order of stability of the protonated amines is greater the size of the ion, lesser is the solvation and lesser is the stability. In case of secondary and tertiary amines, due to steric hindrance, the alkyl groups decrease the number of water molecules that can approach the protonated amine. Therefore the order of basicity is,



Based on these effects we can conclude that the order of basic strength in case of alkyl substituted amines in aqueous solution is

$$(CH_3)_2\ddot{N}H > CH_3\text{-}\ddot{N}H_2 > (CH_3)_3\ddot{N} > \ddot{N}H_3$$

$$(C_2H_5)_2\ddot{N}H > (C_2H_5)_3\ddot{N} > C_2H_5\ddot{N}H_2 > \ddot{N}H_3$$

The resultant of +I effect, steric effect and hydration effect cause the 2^0 amine, more basic.

Basic strength of aniline

In aniline, the NH_2 group is directly attached to the benzene ring. The lone pair of electron on nitrogen atom in aniline gets delocalised over the benzene ring and hence it is less available for protonation makes the, aromatic amines (aniline) less basic than NH_3.

In case of substituted aniline, electron releasing groups like $-CH_3$,$-OCH_3$,$-NH_2$ increase the basic strength and electron withdrawing group like $-NO_2$,-X,-COOH decrease the basic strength.

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